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Non-Coherent Charge Transport in Donor−Acceptor Systems: A Self-Consistent Description of the Intramolecular Charge Flow

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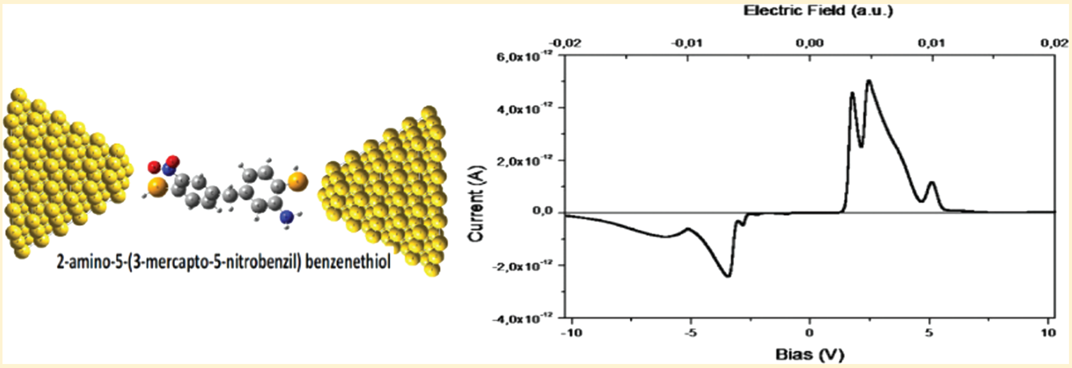
Supporting Information

ABSTRACT:



We have investigated the role that molecular orbitals (MOs) play in the electron transport through a single donor−acceptor molecule submitted to an external voltage applied through two metallic electrodes. Considering the weak coupling limit, in which level-broadening effects are negligible, we investigate the microscopic processes associated to the charge flow through the molecule by examining how the individual molecular levels actually respond to the external electric field. By taking into account the active role played by the MOs in the charge transport between the two electrodes, we have shown that an important contribution may arise in the situation of a field-induced "avoided-crossing" between neighboring energy levels, especially if the corresponding MOs are localized in different regions of the molecule. Conduction channels can be opened or closed as the result of "avoided-crossing" situations in which the spatial localization of the MOs considered changes between the acceptor and the donor opposite sides of the molecule. Our results indicate that the charge transfer between the electrodes is mainly dominated by noncoherent mechanisms involving hole transport through the uppermost occupied molecular orbitals. We suggest that these field-induced changes in the molecular environment may play a key role in the overall transport process and should be considered whenever actual measurements are being performed in single molecules.

1. INTRODUCTION

There is a growing consensus that within the next two decades the continuous scale-down of the physical features of electronic circuits observed in the last 40 years will reach its final limits, both for economic and technical reasons.[1] Among the possible substitutes of the silicon-based microelectronics, molecular electronics (molectronics) appears as an interesting alternative.[2] As a current frontier of knowledge in the interface between physics, chemistry, biology, and electronics at the nanoscopic scale, molectronics offers multiple appeals: while from a scientific perspective there is the fascinating challenge of predicting how single molecules (or a limited number of them) can control the storage and flow of information, from the vantage point of more practical applications there is the ever-elusive goal of reproducing the inner workings of biological systems in a man-made setup.

The first modern proposal of a molectronics device was presented in 1974 by Aviram and Ratner[3] who suggested that molecules composed by an electron donor (D) species connected to an electron acceptor (A) group through an (even short) aliphatic bridge (σ) could act as current rectifiers. They have shown that in this case the electron transport occurs by direct electrode−electrode tunneling where the asymmetric structure of the D−σ−A system responds for the resulting preferential directionality of the current. In the past decade several examples of similar molecular diodes, where the unidirectional intramolecular electron transfer corresponds to a key step in the overall current flow, have been discussed.[4,5]

Mizuseki et al., for instance, have analyzed molecules where D−A groups are connected via an aliphatic chain that acts like a spacer and provides a potential barrier for the electron transport from one end to other.[6] It is assumed that the unoccupied orbitals serve as channels for the flow of electrons through the molecules,

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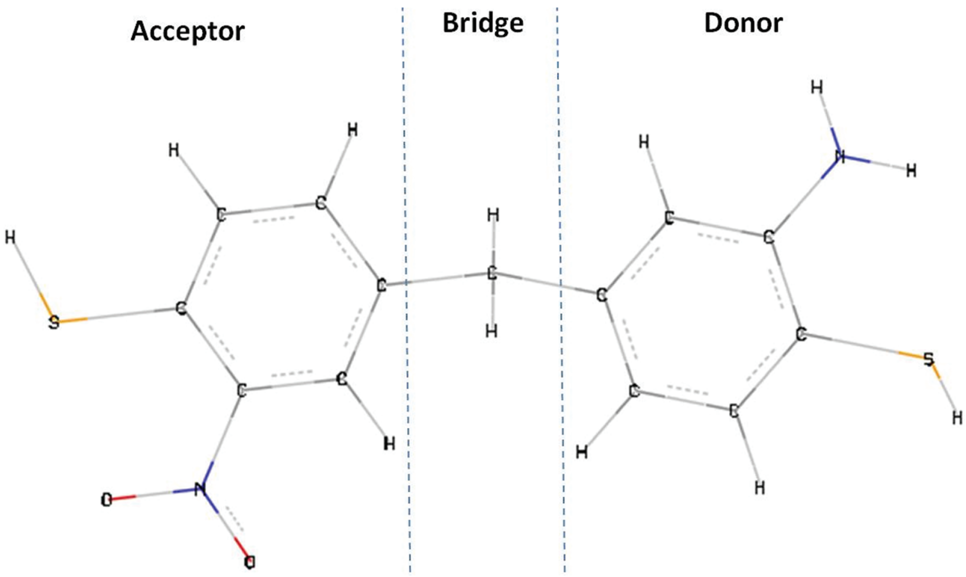

Figure 1. In the AMNB molecule, the acceptor group (A; left) connected to the donor group (D; right) through a saturated bridge (σ, middle).

while the potential drop is considered as the difference in energy between the lowest unoccupied orbitals localized on opposite (donor and acceptor) sides of the molecule.[6]

An alternative point of view for dealing with the problem corresponds to the transmission formalism (or Landauer approach),[7] in which the two metallic electrodes are assumed to be connected by well-defined channel subbands, each one having a specific energy-momentum relationship. In that example of coherent transport formalism, the molecule is considered essentially as a passive element whose characteristics are not affected by the presence of the applied external field or the interaction with the contacts.

Electron motion through a channel, however, is usually affected by "phase-breaking" processes, which arise from the interaction of electrons with photons, phonons and other electrons, giving origin to a noncoherent transport.[7] In fact, electronic correlations must be favored in systems of reduced sizes, such as molecular or quantum dot devices, since lower dimensions and/or very dilute limits are associated to poor screening of the electron−electron interaction, and therefore to an enhancement of the role of Coulomb repulsions.[8]

In this work, we will focus in the case of (charge effects related) noncoherent transport in a given molecular device. For this purpose, we will consider that the molecule−electrode interaction is feeble, so that we are in the so-called weak coupling (also referred to as Coulomb blockade) regime,[9] where broadening effects of the molecular levels can be neglected.[7] Hence, as a first approximation, we can consider that the molecule is not covalently coupled to the electrodes (see more on this later). Instead, we will take into special account the electron−electron interactions in the molecular region and stress the importance of treating molecules not as structureless quantum dots but as more complex systems with active components that not only exchange charge with the two metallic contacts attached to their extremities but effectively participate of the overall electronic flow that passes through them, so that the molecular levels are self-consistently adjusted in response to the externally applied electric field. In particular, we call attention to the fact that the crossing of neighboring molecular levels at certain specific values of the field intensity can result in a sudden rearrangement of the spatial localization of the internal molecular electronic density, in a manner that could dramatically alter the electron flow to and from the adjoining metallic contacts. As a consequence of these localization changes, conduction channels can be opened or closed, either substantially increasing or otherwise hindering the overall charge transport.

As a convenient example of this approach, we have chosen to examine the molecule similar to the one previously studied by Majumder and collaborators,[10] 2-amino-5-(3-mercapto-5-nitro-benzil) benzenethiol (AMNB), with both ends capped by a thiol (−S−H) group (see Figure 1). AMNB is a D−σ−A system compounded by two benzene rings with a donor functional group (amine) and a acceptor functional group (nitrobenzene), joined by a methylene bridge. The thiol groups have been included because of our desire to compare the final conductance results to available experimental data. Due to their high electron affinity for Au electrodes,[11] thiol groups are known to act as convenient molecular clamps (Figure 1), when the hydrogen atoms are lost after the S−Au thiol bond is formed, leading to the self-consistent field (or strong coupling) regime.[7,9,12] However, consistent with the weak coupling assumption, we will consider here the physisorption limit[13] that is valid up to temperatures slightly below 300 K, so that no direct S−Au covalent bond is formed. (In fact, even when true Au−S−H thiolate bonds[12,14] are formed the weak coupling limit would be observed.)

2. COMPUTATIONAL DETAILS

We have adopted the Gaussian 03 program[15] to describe the electronic structure of the AMNB molecule. For this, we first determined its most stable field-free geometrical conformation by use of a 6-31G(d,p) basis function at the B3LYP approximation,[16] a density functional level of treatment. The inclusion of polarization functions is important to improve the description of the virtual orbitals, which may play a relevant role in the electron transfer through the molecule.[17] Then, keeping the optimized geometry frozen, we have determined the total energy in the presence of an external electric field applied along the z-direction

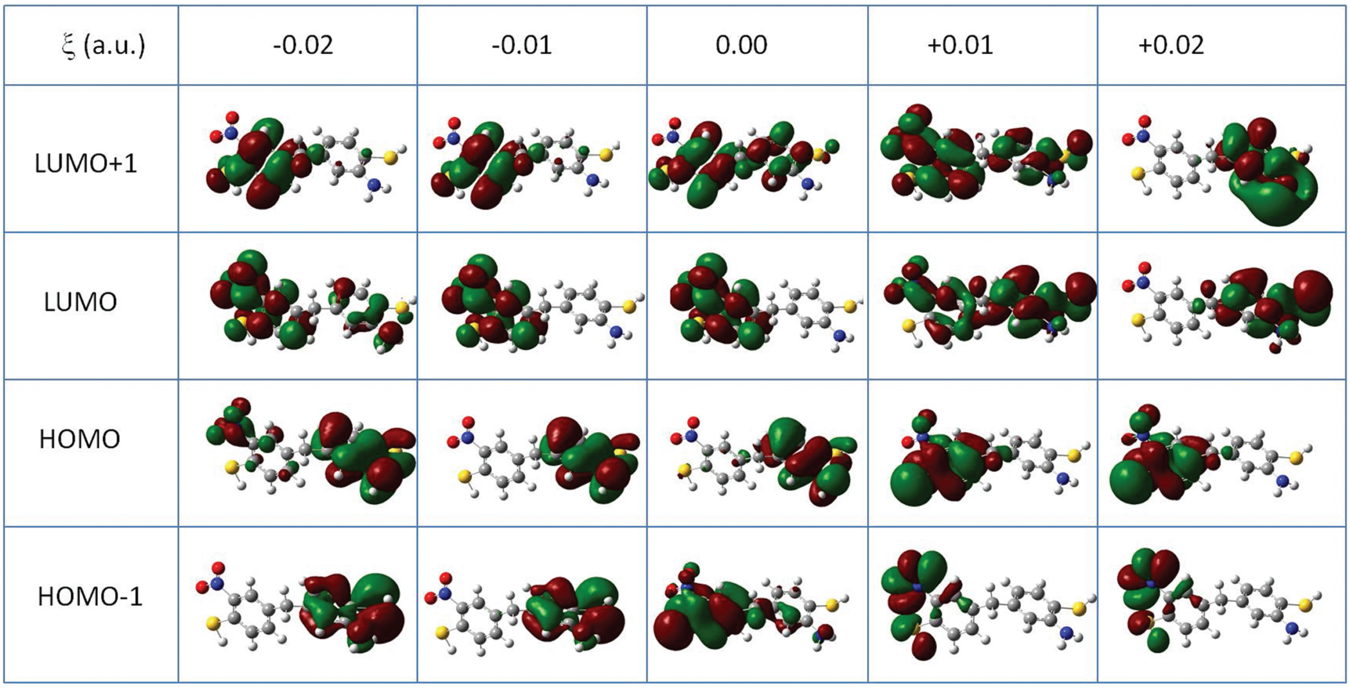

Figure 2. Density of probability of frontier molecular orbitals localized in different regions of the molecule when the value ξ of the external electric field is varied.

(which is always chosen as that of the molecular dipole moment) with varying positive and negative intensities. In this manner we assume that the charge transfer process through the molecule occurs in a time scale shorter than that needed for the geometrical reorganization of the molecular system;[18] the electronic reorganization must follow the electron [hole] transfer step from one side of the molecule to the cathode [the anode to the other terminal end of the molecule]. In fact, simple "back of the envelope" calculations of the molecular traversal time, either directly based on the uncertainty principle or on order of magnitudes estimates of the relevant physical parameters, suggest that the characteristic times involved are in the 10^{-1} to 1 fs range. Typical time scales of single-molecule charge transport events are therefore shorter than those associated to nuclear motion or phonon inelastic effects.[19]

3. RESULTS

Many authors[3,20] have already pointed out that in the strong coupling regime the spatial localization of the frontier molecular orbitals (FMOs) plays an essential role in regulating the charge transport along a given molecule; however, in the weak coupling limit—where transport is supposed to occur by sequential tunneling—the molecule is commonly treated as a structureless quantum dot whose molecular orbitals are frozen no matter how intense is the applied external field. For example, an electron can tunnel from one metallic contact to the lowest unoccupied molecular orbital (LUMO) only if this FMO has a probability of being found "near" this electrode. In the same way, an electron can escape from the highest occupied molecular orbital (HOMO) to the other contact only if a considerable fraction of its density of probability is located in the spatial region of the metallic tip. As a general rule, rectification is more likely to occur in organic devices if the FMOs are localized in a favorable manner (i.e., unoccupied orbitals near the cathode, while the occupied ones are close to the anode). As a consequence, we will argue that, if the spatial localization of a pair of FMOs is somehow exchanged (for instance, as a result of an "avoided-crossing" type of effect[21] due to the applied electric field), conductance channels can be suddenly opened or closed at specific field intensities. We will show in a moment that with this consideration finer details usually seen in experimental $I \times V$ curves,[22] such as negative-differential resistance (NDR),[23] can be easily explained. Therefore, the quantification of the degree of localization of a given molecular orbital is an important parameter for the understanding of transport processes in molecular systems.

Spatial Localization of the Frontier Molecular Orbitals: Qualitative Aspects. When no external electric field is present, it is well-known that the HOMO (LUMO) of D$-\sigma-$A systems is spatially localized mainly in the donor (acceptor) side of the molecule, as Majumder and collaborators have shown to occur for the AMNB molecule.[10] However, the localization rule valid for the HOMO and LUMO cannot be extended to describe the spatial distribution of other FMOs in general. For instance, as it can be seen from the probability isosurfaces depicted in Figure 2, while the HOMO-1 is mainly localized in the acceptor part of the AMNB molecule, the LUMO+1 is delocalized through the whole system.

We have found that the presence of an external electric field alters the spatial localization of the FMOs in the sense that, when it is applied in the $-\hat{z}$ direction (i.e., oriented from D to A; please note that the AMNB molecule does not exhibit necessarily a planar configuration), the occupied [unoccupied] molecular orbitals tend to concentrate in the donor [acceptor] end of the molecule, and that this localization is prone to be reversed when the field is applied in the opposite (i.e., $+\hat{z}$) direction. For example, in the case of the AMNB molecule we show in Figure 2 that when an external field of $\xi_z = -0.02$ au is applied, both

Table 1. Degree of Localization of the HOMO-1, HOMO, LUMO, and LUMO+1 Frontier Orbitals in the Donor ($\Gamma_D{}^0$), Bridge ($\Gamma_B{}^0$), and Acceptor ($\Gamma_A{}^0$) Parts of the Molecule for Some Selected Values of the Applied Electric Field

	HOMO-1			HOMO			LUMO			LUMO+1		
ξ (au)	$\Gamma_D{}^0$	$\Gamma_B{}^0$	$\Gamma_A{}^0$	$\Gamma_D{}^0$	$\Gamma_B{}^0$	$\Gamma_A{}^0$	$\Gamma_D{}^0$	$\Gamma_B{}^0$	$\Gamma_A{}^0$	$\Gamma_D{}^0$	$\Gamma_B{}^0$	$\Gamma_A{}^0$
−0.02	0.998	0.001	0.001	0.872	0.001	0.127	0.126	0.000	0.874	0.038	0.017	0.945
−0.01	0.994	0.003	0.003	0.993	0.001	0.006	0.004	0.000	0.997	0.040	0.010	0.950
0.00	0.074	0.000	0.926	0.979	0.003	0.018	0.004	0.002	0.996	0.219	0.011	0.770
0.01	0.002	0.004	0.994	0.026	0.009	0.965	0.749	0.006	0.245	0.247	0.015	0.738
0.02	0.000	0.000	1.000	0.017	0.003	0.980	0.990	0.005	0.005	0.990	0.007	0.003

HOMO and HOMO-1 [LUMO and LUMO+1] are in fact mainly localized in the donor [acceptor] region. A complete reversal of the spatial localization of these FMOs is observed for an applied field of ξ_z = +0.02 au. We will introduce below a simple recipe for expressing these results in a more quantitative manner.

Spatial Localization of the Frontier Molecular Orbitals: Quantitative Aspects. To establish the criterion of spatial localization on a more firm basis, when the molecule (M) is in a S-charge state we will divide the atomic orbitals $|\phi_\mu\rangle$ as belonging to three contiguous regions, namely, the donor group (D), the bridge (B), and the acceptor group (A),[24] so that any molecular orbital could be written as

$$|\psi_i^S\rangle = \sum_\mu^M c_{\mu i}^S|\varphi_\mu\rangle$$
$$= \sum_\mu^D c_{\mu i}^S|\varphi_\mu\rangle + \sum_\mu^B c_{\mu i}^S|\varphi_\mu\rangle + \sum_\mu^A c_{\mu i}^S|\varphi_\mu\rangle \tag{1}$$

As the molecular orbitals are orthonormal, the relationship

$$\langle\psi_i^S|\psi_j^S\rangle = \delta_{ij} = \Gamma_{i,D}^S + \Gamma_{i,B}^S + \Gamma_{i,A}^S \tag{2}$$

is automatically satisfied, provided that

$$\Gamma_{i,D}^S = \sum_\mu^D \sum_\nu^A (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^D \sum_\nu^B (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^D \sum_\nu^D (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S \tag{3}$$

$$\Gamma_{i,B}^S = \sum_\mu^B \sum_\nu^A (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^B \sum_\nu^B (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^B \sum_\nu^D (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S \tag{4}$$

and

$$\Gamma_{i,A}^S = \sum_\mu^A \sum_\nu^A (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^A \sum_\nu^B (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S + \sum_\mu^A \sum_\nu^D (c_{\nu i}^S)^* S_{\nu\mu} c_{\mu i}^S \tag{5}$$

As usual,[24b] one can interpret the above relations by considering that $\Gamma_{i,D}^S$, $\Gamma_{i,B}^S$, and $\Gamma_{i,A}^S$ correspond to the probability of finding the $|\phi_i^s\rangle$ molecular orbital localized in the D, B, and A regions, respectively. With this, a simple quantitative measurement of the spatial localization of the MOs becomes available,[24] and we can put the preceding qualitative discussion of how the external field affects the electronic distribution of the AMNB molecule on a numerical comparative basis. (As we will see below, the important point here is that the presence of the saturated bridge favors the spatial localization of the HOMO and the LUMO on different sides of the molecule.)

In Table 1 we present the gradual change in the spatial localization of the FMOs of the neutral (i.e., S = 0) AMNB molecule as the intensity of the applied external field progressively changes from −0.02 au to +0.02 au.

Let us first consider the zero-field case, when no external perturbation is applied. As expected, the HOMO [LUMO] is essentially localized in the donor [acceptor] region, with a $\Gamma^0_{\text{HOMO,D}}$ = 97.9% [$\Gamma^0_{\text{LUMO,A}}$ = 99.6%]. As for the other FMOs, one can see that the localization character is different, as qualitatively seen in Figure 2: while the HOMO-1 is mainly localized in the acceptor side of the molecule ($\Gamma^0_{\text{HOMO-1,D}}$ is only 7.4%), the LUMO+1 has a stronger acceptor characteristic ($\Gamma^0_{\text{LUMO+1,A}}$ = 77.0%). Note that these orbitals have a negligible probability of localization in the bridge region. The identification of these differences in the spatial localization of the pairs (HOMO and HOMO-1) and (LUMO and LUMO+1) is the basis for our anticipation that an interesting behavior could arise if these pairs of neighboring FMOs somehow exchange their localization characteristics.

A possible way to induce this is by turning on the external field along the axis connecting the two metallic terminals. At one side, the Fermi level μ_1 of the cathode, which varies linearly with the field strength, will eventually cross the energy of successive unoccupied molecular orbitals; at the other, a similar—but symmetrically opposed—variation of the Fermi level μ_2 of the anode will cause the crossing with successive occupied MOs. If the FMOs are differently affected by their interaction with the field of intensity ξ, then the charge transport characteristics along the molecule, as measured for example by its conductance, could be very sensitive to the value of ξ.

This is in fact the case, as even a cursory analysis of Figure 2 will reveal. If we consider the ξ > 0 case, for instance, we see that for ξ = 0.01 au a situation of transition in the localization of the FMOs seems to be observed: while the HOMO [HOMO-1] is already essentially localized in the acceptor region, the LUMO and LUMO+1 are still delocalized. However, with a further increase in the field intensity (ξ = 0.02 au), the HOMO-1 and HOMO remain essentially localized in the acceptor extremity of the molecule, but the two lowest UMOs (LUMO and LUMO+1) now become confined to the donor moieties. In all cases examined, the probability of localization of the FMOs in the saturated bridge is essentially zero, that is, $\Gamma^0_{i,B} \cong 0$.

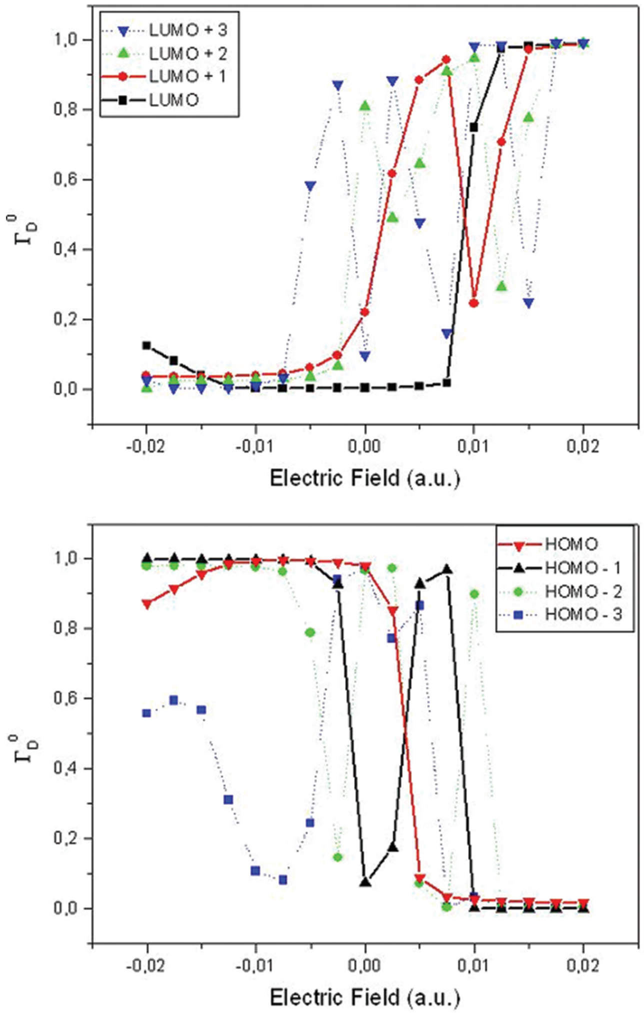

Figure 3. Degree of spatial localization in the donor region (Γ_D), which varies as a function of the value ξ of the external electric field, as it can be seen for the case of UMOs (top) and OMOs (bottom); special emphasis is given to the Γ_D variation of the pairs (HOMO, HOMO-1) and (LUMO, LUMO+1).

We will use the ξ-dependence of the $\Gamma^0_{i,D}$ and $\Gamma^0_{i,A}$ matrices to represent how the localization of the FMOs in the donor and acceptor regions will evolve as a function of the strength of the applied field. In Figure 3, we plot the corresponding probabilities of finding the occupied and unoccupied MOs in these regions for different values of ξ. Naturally, $\Gamma^0_{i,D} + \Gamma^0_{i,A} \cong 1$ for each orbital, if we consider the vanishingly small value of $\Gamma^0_{i,B}$. For the $\xi < 0$ case, the occupied MOs are localized in the donor region, with the exception of the HOMO-3, whose localization alternates between the D and A regions. However, for field intensities in the -0.01 au $\leq \xi \leq 0.01$ au range of values, we find a "transition" region, where the spatial localization of the occupied MOs moves to the acceptor side of the molecule. This transition can happen in a smooth manner, as it is the HOMO case, or as a series of sudden changes (as observed for the HOMO-i, i = 1, 2, and 3), where $\Gamma^0_{i,D}$ and $\Gamma^0_{i,A}$ alternate values in a rapid succession. As for the unoccupied MOs, a similar trend can be identified, in that these FMOs go from an initial localization (for -0.02 au $\leq \xi \leq$ -0.01 au) entirely in the acceptor side of the molecule to a final situation (for $\xi > 0$) where they become concentrated in the donor moiety of the system. Here the transition region seems to occur in the (-0.005 au $\leq \xi \leq 0.014$ au) interval.

Let us now examine the behavior of the FMOs in these transition regions in a more detailed manner. As an example, we can compare the curves of Figure 3 top and bottom, where in each case we highlight the behavior of $\Gamma^0_{i,D}(\xi)$ for the two highest and the two lowest MOs, respectively. For -0.02 au $\leq \xi \leq$ -0.01 au, these occupied [unoccupied] MOs are localized mainly in the donor [acceptor] side of the molecules, while for the largest positive values of ξ they become localized in the opposite side, i.e., near the acceptor [donor] groups, a set of results that can be easily interpreted. However, at intermediary values of the field applied in the positive direction, a crossing is seen to occur in the values of $\Gamma^0_{i,D}(\xi)$ for both pairs of MOs; this can be seen for the HOMO-1 and HOMO pair in the 0 au < ξ < 0.0075 au range (Figure 3, bottom), and for the LUMO and LUMO+1 pair (Figure 3, top) in the 0.005 au < ξ < 0.01 au interval. At each of these crossings, the spatial localization of the orbitals involved changes in a very appreciable manner within a relatively small range of variation of the intensity of the applied field.

As a general trend we can observe that for large values of $|\xi|$, when the field is applied along the negative direction, the occupied [unoccupied] MOs become localized in the donor [acceptor] side of the molecule; however, each of these orbitals become localized in the opposite manner for an applied field of same strength but inverse direction. In addition, as shown in Figure 3, for intermediary values of $|\xi|$ there is a limited set of windows of values for which the spatial localization of the four higher occupied and the four lower unoccupied MOs oscillate between the opposite sides of the molecule, in rapid succession.

Importance of the Field-Induced Mixing of Orbitals. The change in the spatial localization of the molecular orbitals associated to energy levels that are involved in an "avoided-crossing" situation can be easily understood by considering two levels $|\varphi_1\rangle$ and $|\varphi_2\rangle$ and their corresponding eigenvalues E_1 and E_2 associated to a nonperturbed Hamiltonian $\mathbf{H}_0$.[25] In the presence of an external electric field $\vec{\xi}$ applied along the $\hat{z}$-direction, the Hamiltonian can be written as $\mathbf{H} = \mathbf{H}_0 + \mathbf{W}$, where $|\mathbf{W}| = ez\xi$. The eigenstates and eigenvalues of $\mathbf{H}$ are given respectively by

$$\begin{pmatrix} |\chi_+\rangle \\ |\chi_-\rangle \end{pmatrix} = \begin{pmatrix} +\cos\left(\dfrac{\theta}{2}\right) \times e^{-i\delta/2} & \sin\left(\dfrac{\theta}{2}\right) \times e^{+i\delta/2} \\ -\sin\left(\dfrac{\theta}{2}\right) \times e^{-i\delta/2} & \cos\left(\dfrac{\theta}{2}\right) \times e^{+i\delta/2} \end{pmatrix} \begin{pmatrix} |\varphi_1\rangle \\ |\varphi_2\rangle \end{pmatrix} \tag{6}$$

and

$$E_\pm = \frac{1}{2}(\Delta_1 + \Delta_2) \pm \frac{1}{2}\sqrt{(\Delta_1 - \Delta_2)^2 + 4|W_{12}|^2} \tag{7}$$

where δ is a phase factor, $\Delta_1 = E_1 + W_{11}$, $\Delta_2 = E_2 + W_{22}$, and $\tan\theta = ((2|W_{12}|)/(\Delta_1 - \Delta_2))$ $(0 \leq \theta \leq \pi)$.

Then, we see that, by the effect of the interaction W, each one of the perturbed eigenstates correspond to a mixing of the two pure original $|\varphi_1\rangle$ and $|\varphi_2\rangle$ states. However, an exchange between the characteristics of the two original states will occur, in particular when the mixing parameter changes from its initial value ($\theta = 0$, i.e., well before the interaction is switched on) to its final limit ($\theta = \pi$, i.e., long afterward it has been switched off) situation. In this manner, at an avoided-crossing situation between two levels $|\varphi_1\rangle$ and $|\varphi_2\rangle$ that are localized in different parts of the molecule (as, for example, in the donor and acceptor regions), the applied electric field can dramatically modify the electronic distribution of the system considered. Hence, the avoided-crossing phenomenon may introduce a very peculiar situation for the overall electronic flow, since at specific values of the applied electric field strength two neighboring levels that did not contribute before to the process can suddenly become important channels of charge transport.

Effect of the Orbital Mixing on the Noncoherent Electron Transport. Consider, for example, an Aviram and Ratner system composed by an acceptor−(σ-bridge)−donor molecule physisorbed to two electrodes in such a way that the acceptor part is connected to the left lead (M1) and the donor part to the right lead (M2). An electron can tunnel from the left electrode to the LUMO only if this molecular orbital has a finite probability of being localized in the acceptor part of the molecule and provided that this level is inside the corresponding Fermi window (which in here will be understood as the energy region defined by the Fermi level mismatch $|\mu_2 - \mu_1|$ between the M1 and M2 electrodes) at the specific field intensity.[26]

However, we can imagine a situation where the electron can be transferred from the cathode to a molecule even when the LUMO is not initially localized in the acceptor region, provided that this spatial localization condition is satisfied by the LUMO+1. Once this requirement is satisfied, it suffices that at a given range of the applied field's intensity an avoided crossing situation develops between the LUMO and LUMO+1 within the Fermi window of the problem. As discussed above (see eq 6), as the applied field increases in intensity, the original character of the two levels involved in the avoided crossing will mix, and as a consequence, the localization of the LUMO in the acceptor region can become large enough to favor the electron transfer from the cathode.

Naturally, the reverse situation can also occur. In this manner, even if the LUMO enters the Fermi window while it is originally localized in the acceptor part of the molecule, the transfer of charge can be hampered if at the same energy range an avoided crossing situation develops between the LUMO and LUMO+1, while the spatial localization of the latter mainly occurs far from the acceptor region. Now, there will be a critical value of the applied field strength above which the electron transfer from the cathode will dramatically decrease. One can then expect that the corresponding $I \times V$ curve will present NDR characteristics; that is, in spite of the increasing applied voltage a peak will arise in the electrical response due to a subsequent current decrease once one specific electron transfer channel is closed.

We are now in a better situation to understand the overall behavior of the electron transfer between the molecule and the connecting electrodes, when the intensity ξ of the applied electric field is progressively increased. As it can be easily identified in Figure 4, where we plot the variation of the Fermi level of the metallic contacts and of the molecular energy levels as functions of ξ, it is exactly for the "transition regions" of certain MOs, when their spatial localization $\Gamma^0_{i,\mathrm{D}}(\xi)$ changes substantially (and sometimes in an alternating pattern), that one can identify examples of "avoided-crossing" for the corresponding pairs of energy levels. For instance, we have shown before that for 0.002 au < ξ < 0.008 au the spatial localization in the donor region of the HOMO and HOMO-1 ($\Gamma^0_{i,\mathrm{D}}(\xi)$) changes in a complementary way (see bottom panel of Figure 3), and that a similar pattern is found for the $\Gamma^0_{i,\mathrm{D}}(\xi)$ of the LUMO and LUMO+1 pair (top panel of Figure 3), for 0.006 au < ξ < 0.010 au. Exactly at

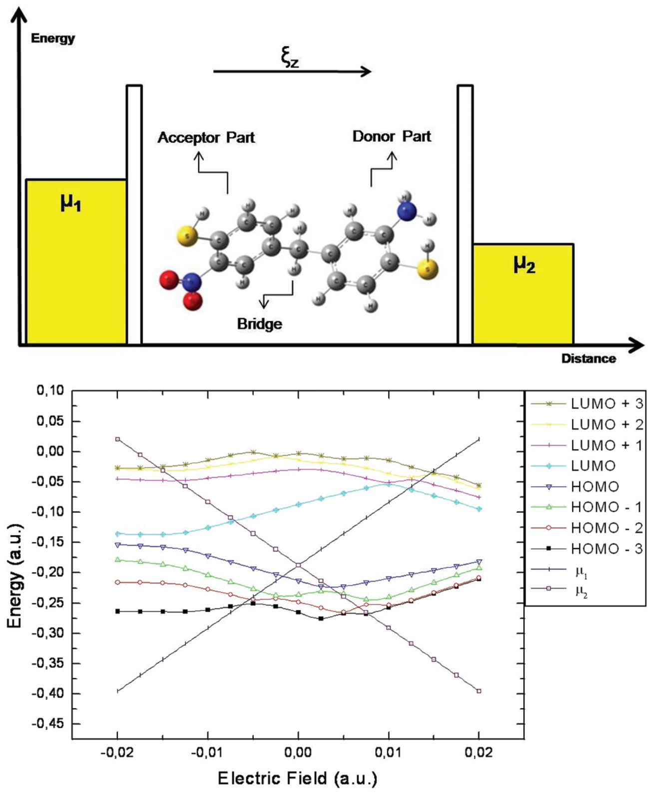

Figure 4. Top: diagram representing the AMNB molecule connected to two gold electrodes; the presence of each thiol group is modeled by a rectangular barrier. Bottom: variation of the energy levels of some frontier molecular orbitals as a function of ξ, the value of the external electric field.

these two ranges of the applied field intensity one can identify in Figure 4 the existence of the avoided-crossing phenomenon for the corresponding pair of FMOs. Other regions where a similar pattern of avoided-crossing between neighboring MOs exists can be shown to correspond to situations of dramatic change in the spatial localization of the orbitals involved.

Microscopic Charge Transfer Mechanisms. To describe the flow of current through the molecule and, as a consequence, the conductance of the entire system, we have considered alternative sequential three-step processes.

In the first possible mechanism, the initial charge transfer involves an electron being transferred from the left electrode (M1) to the molecule:

$$\mathrm{M1} \overset{-e}{\underset{}{\rightleftharpoons}} \mathrm{AMNB}^- - M2 \rightarrow \mathrm{M1} - AMNB^* \overset{-e}{\underset{}{\rightleftharpoons}} \mathrm{M2} \rightarrow \mathrm{M1} - \mathrm{AMNB} - \mathrm{M2} \qquad (9a)$$

In this "electron transfer" mode, we consider that: (i) an electron leaves the cathode (M1) and enters the acceptor region of the AMNB molecule (which becomes negatively charged), (ii) an electron leaves the donor region of the AMNB$^-$ anion (and the molecule returns to its neutral form, albeit in an excited state) and is transferred to the anode (M2), and (iii) the AMNB* molecule would then relax to the ground AMNB neutral state. The i−iii cycle repeats itself.

Another possibility is that the initial charge transfer involves a hole being transferred from the right electrode (M2) to the molecule, which is equivalent to say that an electron leaves the molecule and enters the unoccupied Fermi region of the metal:

$$\mathrm{M1} - AMNB^{+} \xleftarrow[-]{+e} \mathrm{M2} \rightarrow \mathrm{M1} \xleftarrow[-]{+e} \mathrm{AMNB}^{*} - M2 \rightarrow \mathrm{M1} - AMNB - M2 \qquad (9b)$$

In this "hole transfer" mode, we consider that: (i) an electron leaves the acceptor region of the neutral AMNB molecule (which then becomes positively charged) and enters the anode (M2), (ii) an electron leaves the cathode (M1) and enters the donor region of the $AMNB^+$ cation (and the molecule returns to its neutral form, albeit in an excited state), and (iii) the AMNB* molecule would then relax to the ground AMNB neutral state. The i–iii cycle repeats itself.

In each step, the minimum energy configuration of the molecule—be it in its neutral ($S = 0$) form (both in the excited as in the ground state) or as a single charged species (both as an anion, when $S = -1$, or a cation, when $S = +1$)—is fully considered. However, since we assume that the electron transfer processes occur within a shorter time scale than that involved in the nuclear reorganization, in all cases we consider the ground state geometry of the neutral AMNB species.

Current and Conductance. As a reasonable first approximation, we will consider the weakly coupled regime for which no ballistic (i.e., coherent) process exists and the molecule can be treated as an isolated (finite) well, so far from the electrodes that its levels do not interact in a significant extent with their evanescent states; hence, no broadening of the molecular levels[7] needs to be considered as for each charge state of the AMNB molecule the corresponding molecular states remain stationary into the well.

As usual in the literature,[27] we can assume that in each case the three microscopic charge transfer steps involved in the passage of the electric current occur in an independent manner, so that the probability P for the overall charge transport throughout the entire system (i.e., between the two M1 and M2 electrodes) can be described as a product of the three independent probabilities: P_{in}, which is related to the electrode–molecule charge transfer (when the molecule becomes charged), P_{out}, which corresponds to the molecule–electrode charge transfer (the molecule returns to its neutral form albeit in an excited state), and $T_{\sigma\lambda}$, the probability that the final intramolecular electronic relaxation (when the neutral molecule returns to its fundamental state) takes place. However, as in the multielectron master equation picture,[7,9,28] we can consider that the first two of these probabilities (i.e., those related to the left/right electrode–molecule transfer) may take into account the possibility of different charge states that correspond to each one of the separate processes involved.

The most simple and straightforward manner of doing this is by considering eq 10 that expresses in a general form[24a] the total probability P as a product of the isolated probabilities $P_{\mathrm{in}}^{\sigma S}$ that an electron (see eq 9a) [hole (see eq 9b)] is transferred from the electrode to the σ OMO of the molecule times $P_{\mathrm{out}}^{\lambda S}$, the probability that it leaves the molecule through an available λ UMO toward the opposite electrode, times the probability $T_{\sigma_0\lambda_0}$ that a UMO–OMO transition occurs as the final step, that is,

$$P = \sum_{\lambda}^{n} \sum_{\sigma}^{n} \sum_{X=0}^{1} (P_{\mathrm{in1}}^{\sigma_X} + iP_{\mathrm{in2}}^{\sigma_X})(T_{\sigma_0\lambda_0})(P_{\mathrm{out2}}^{\lambda_{X-1}} + iP_{\mathrm{out1}}^{\lambda_{X-1}}) \qquad (10)$$

where the subscripts 1 and 2 represent the left [right] electrode, in [out][29] indicates charge transfer to [from][5] the molecule, the sum in σ and λ runs over the relevant FMOs, and the indexes $S = X$ and $S = X - 1$ are integers related to the possible charged states in which the molecule can be found during the electron flow process (naturally, $X = \{0,1\}$ for the transfer of a single charge).

In this compact notation, P can be thought as associated to a generalized conductance, since $\mathrm{Re}(P)$ represents the net probability (i.e., M1→M2 minus M2→M1) of charge flow (a dispersive-type contribution to the current associated to the motion of free charges) throughout the device, while the $\mathrm{Im}(P)$ term can be interpreted as a molecular Maxwell displacement current probability (a type of charge storage term, when charge is transferred back-and-forth between a given side of the molecule and its neighboring electrode). Note that in the above expression it is considered that two independent electrons are transferred at the two junctions, with the molecule being in a different charge state on each step. This would correspond to a noncoherent mechanism of transport associated to charging effects, a process also known as cotunneling,[29] and the charged states can be viewed as virtual states that are only occupied during the process. Although eq 10 could be generalized to include ballistic transport, this will not be done here since we are focusing in the cotunneling regime because of the σ-character of the bridge connecting the D and A groups in the AMNB molecule.

In the present work, where we are interested in the current passing through the device, we will center our analysis in the real part of the probability only. The numerical value of the probability $P_{\mathrm{in/out}}^{\lambda S}$ can be obtained by dividing the probability for a single charge tunneling through the M1-AMNB-M2 system by the normalized flux of incident electrons, that is, $P_{\mathrm{in/out}}^{\lambda S}$ = (tunneling flux)/(incident flux).

The incident flux can be obtained by considering the electrodes as infinite systems (hence, allowing the use of a continuous density of states) while the molecule is described by a discrete density of states (represented by the manifold corresponding to the limited set of FMOs considered).

Expressions for the incident and tunneling flux are presented in many elementary text books (see ref 5, for example). We will adopt a trapezoidal barrier to represent the (physisorbed) sulfur–gold electrode junction and express the density of states of the molecule via a projected density of states (PDOS) corresponding to the density of states of the donor and acceptor sides of the molecule (supposed to be in a S-charge state) as[7]

$$D^S = \sum_i \delta(E - E_{i,S}) = \sum_i \langle\psi_i^S|\psi_i^S\rangle\delta(E - E_{i,S})$$
$$= \sum_i (\Gamma_{i,\mathrm{D}}^S + \Gamma_{i,\mathrm{B}}^S + \Gamma_{i,\mathrm{A}}^S)\delta(E - E_{i,S})$$
$$D^S = D_{\mathrm{D}}^S + D_{\mathrm{B}}^S + D_{\mathrm{A}}^S \qquad (11)$$

where ψ_i^S is the i-th FMO in a S-charge state. We can interpret the product $\Gamma_{i,\mathrm{D}}^0\delta(E - E_{i,S})$ (or $\Gamma_{i,\mathrm{A}}^0\delta(E - E_{i,S})$) as the density of states projected in the spatial region of the molecule where the donor (or acceptor) molecular electronic density (for the S-charge state) has been estimated. Assuming that the molecule is initially in its neutral state, the current is given by $I = I_0P$, where $I_0 = h^{-1}[2e(\mu_2 - \mu_1)]$ is the elementary current per transversal mode for a ballistic conductor[24a,30] and the conductance $C = \mathrm{d}I/\mathrm{d}V$ was determined numerically (see Figure 5).[24a,30] Finally, the probability $T_{\sigma_0\lambda_0}$ can be obtained by a configuration interaction with single excitations (CIS) calculation[31] for the case $\sigma_0 \neq \lambda_0$, from which the coefficients of the wave function for each excitation and the corresponding oscillator strength can be

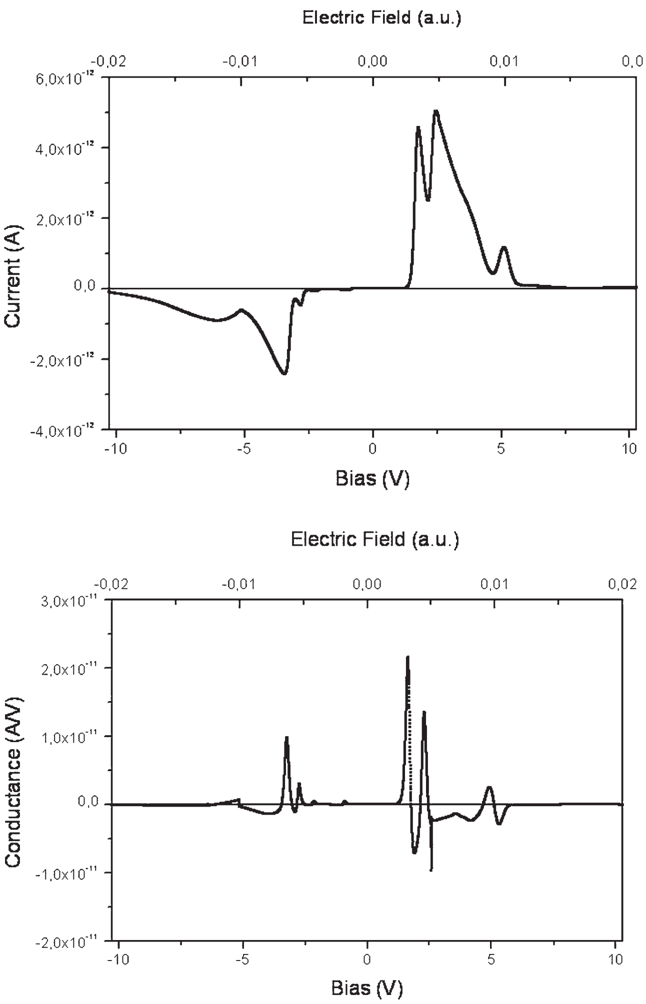

Figure 5. Calculated value of the current (top) through the AMNB molecule and corresponding conductance (bottom) as a function of ξ, the value of the external electric field.

determined by use of the Gaussian 03 program.[15] (Please see additional details concerning the calculation of the total probability P of overall charge transport in the AMNB molecule in the Supporting Information.)

Qualitative Aspects of the ($I \times \xi$) Curve: the Opening and Closing of the Conductive Channels. The key to the understanding of the charge transfer processes is to realize that, because of the linear variation of the Fermi levels μ_1 and μ_2 of the electrodes with the applied field intensity, they begin to cross the energy of individual molecular levels. Naturally, for any charge state S considered, the latter have also their energies readjusted to the new value of the external bias.[32] The transport could occur by transfer of electrons from the OMOs to the electrode Fermi level of lower energy (hole transport), or from the electrode Fermi level to lower energy UMOs (electron transport). However, for this to happen it is essential that the relevant molecular orbitals have a non-negligible spatial localization in the same side of the electrode considered. Since the localization of these orbitals can be dramatically altered by field-effects such as "avoided-crossing" type of events, this is the single most important information to be always taken into consideration. Finally, note that although we mention the crossing of the Fermi level as a definite energy value, at finite temperatures thermal broadening effects will always be present.

One can use the results of Figures 3 and 4 to improve our description of the fine structure (i.e., the presence of characteristic peaks) in the ($I \times \xi$) curve shown in Figure 5. Consider first the case of an electric field applied in the positive region, where one can see that there are two narrow peaks (centered at ξ = 0.003 au and 0.010 au, respectively), plus an intermediary broad band that has a maximum at ξ = 0.0055 au). By combining the information of Figures 3 and 4, one can explain the origin of these features seen in Figure 5 in the following manner:

(a) Peak centered at ξ = 0.003 au: at this value of the applied field strength, the right electrode Fermi level (μ_2) crosses the HOMO level. For this exact value of ξ, the HOMO is mainly localized in the donor side of the molecule ($\Gamma^0_{\text{HOMO,D}} \sim 0.9$; see Figure 3). Hence, in this situation the HOMO level becomes an open channel for hole transport, in agreement with what can be seen in Figure 5. However, for larger values of ξ, the HOMO becomes once again more localized in the acceptor region of the molecule, and this molecular orbital now begins to act as a closed channel, so that a further increase in ξ will lead to a reduction in the value of the calculated current. Note that no other occupied MO can contribute to the charge transport at this point, because so far the right electrode Fermi level has crossed only the highest occupied energy level; then, a typical NDR profile must result.

(b) Broad band centered with a maximum at ξ = 0.005 au: at this value of the applied field, the HOMO is already mainly localized in the acceptor part of the molecule and cannot contribute to the overall (hole) transport. However, μ_2 has already crossed the HOMO-1 level, and at the 0.004 au < ξ < 0.008 au range this occupied MO has $\Gamma^0_{\text{HOMO-1,D}} > 0.9$. As a consequence, the maximum in the measured current decays more slowly, and a broader band must be registered.

(c) Peak centered at ξ = 0.010 au: at this range of values of the applied field strength, both HOMO and HOMO-1 are mainly localized in the acceptor region. However, μ_2 has already crossed the HOMO-2 level, which becomes localized ($\Gamma^0_{\text{HOMO-2,D}} \sim 0.9$) in the donor region exactly at $\xi \sim 0.010$ au.

We have also implemented a confirmatory study (please see the Supporting Information) where the individual contributions of the HOMO, HOMO-1, and HOMO-2 were artificially suppressed from the overall charge transport: in each case, the corresponding peak discussed in the a, b, and c cases above would not be present in the calculated ($I \times \xi$) curve.

In the case of UMOs, although these levels cross the chemical potential (μ_1) of the cathode, only when $\xi \sim 0.008$ au is that their spatial localization becomes mainly confined to the donor region of the AMNB molecule, and therefore the process of charge transfer from the left electrode does not have a high probability of occurring. Consequently, these molecular levels can be considered as permanently closed channels, inaccessible to the overall conduction process.

A similar analysis can be carried for the case of an electric field applied along the negative direction, when the characteristic features of the negative current region curve can be understood.

4. CONCLUSION

In this work we have investigated the role that the molecular orbitals play in the electron transport through an AMNB molecule (a donor−acceptor species) connected to two metallic electrodes submitted to an external voltage. Here, we have considered only the case of noncoherent transport contributions, so that the AMNB molecule could not be treated as a structureless quantum dot. By examining the microscopic processes in which charge is transferred to [from] the molecule from [to] the electrodes, we consider in each case how the individual molecular levels actually respond to the applied electric field. As a consequence, our results are substantially different from those obtained from a coherent transport type of treatment, where the molecule participates of the electron transport only by offering well-defined subbands that act as channels for the electron flow.

While taking into account the active role played by the molecular orbitals in the charge transport between the two electrodes, some authors[33] have considered that the corresponding electronic density necessarily must extend throughout the molecule. While this may be true in the case of coherent transport, we have shown that in the weak coupling limit an important contribution may arise in a situation of field-induced changes in the MOs and "avoided-crossing" between neighboring energy levels, especially if the corresponding MOs are localized in different regions of the molecule. In this manner, we have found that for instance with the increase of the strength of the applied field when the Fermi level of the cathode crosses the energy level of an unoccupied molecular orbital localized in the donor region, a conduction channel can be opened, provided that as the result of a subsequent "avoided-crossing" situation the spatial localization of the MO changes to the acceptor side of the molecule, that is, closer to the other electrode. Naturally, a similar situation can occur in which a conductive channel is opened when the Fermi level of the anode crosses the energy level of an occupied molecular orbital localized in the acceptor region; a subsequent change of the corresponding spatial localization to the donor side of the molecule as a result from an "avoided-crossing" with a neighboring level will result in an increase of the probability of the electron transport between the two electrodes. We believe that the importance of the effects associated to the field-induced changes in the spatial localization of the frontier molecular orbitals effects here discussed have not been properly taken in consideration in different approaches of the charge transport in molecular electronic devices.

Finally, the approximation that the molecular levels are not broadened by their interaction with the electrodes could be improved, for instance by considering an "extended molecule" where small clusters of gold atoms are bound to each end of the AMNB molecule, so that the calculated ab initio molecular levels will now include, from the onset of the calculation, the correction due to the presence of the neighboring electrodes. At the present time, when molectronics enters a new stage of actual measurements being performed in single molecules, the consideration of the molecular entity as an active player in the transport process needs to be pondered in each case.

■ ASSOCIATED CONTENT

S Supporting Information. Additional details concerning the calculation of the total probability *P* of overall charge transport in the AMNB molecule and the role played on it by the HOMO, HOMO-1, and HOMO-2. This material is available free of charge via the Internet at http://pubs.acs.org.

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